# Aditya Mittal

Full Stack Engineer | AI Integration

Noida, Uttar Pradesh, India

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## EDUCATION

**Jaypee University of Information Technology** **2023 – 2027**

*B.Tech – Computer Science and Engineering (Minor: ECE)* *Solan, Himachal Pradesh*

## EXPERIENCE

**Basepair** **June 2026 – August 2026**

*Software Development Engineer in Test (SDET) Intern* *Remote*

- Diagnosed and resolved critical production distributed-systems defects, including an **OAuth race condition** via DB-level row locking and multi-instance file-access failures via **AWS S3** on-demand fallbacks.
- Led a full-stack **SonarQube remediation** initiative across Python/Django and React codebases, resolving 100+ BLOCKER/HIGH security and reliability defects (SQL injection, ReDoS, insecure crypto, unhandled exceptions) and authoring organizational **Standards.md** guidelines.

**Spelll Production** **June 2024 – July 2024**

*Frontend Engineering Intern* *Remote*

- Owned the end-to-end frontend lifecycle from prototyping to production deployment, translating UI/UX mockups into pixel-perfect, mobile-first components and integrating client-side form validation to improve engagement and lead conversion.

## PROJECTS

**MedVault** | Django, DRF, PostgreSQL, Gemini API, React, JWT | Live Demo **Feb 2026 – Aug 2026**

- Engineered a structured **Google Gemini API prompting pipeline** with a hard-coded JSON output schema and explicit safety constraints preventing diagnosis/prescription hallucinations, capping input at 50,000 characters and adding a tolerant **JSON-from-prose fallback parser** for malformed model output.
- Designed a **Django REST Framework** backend with a custom email-based User model, **JWT authentication with refresh-token rotation**, and per-user data isolation enforced at the queryset level; isolated all Gemini calls behind a single mockable client so a **47-test backend suite** runs with zero API key configured.

**Canopy** | Django, PostgreSQL, tree-sitter, Gemini API, Plotly Dash | Live Demo **Aug 2026 – Sep 2026**

- Designed a **multi-language static-analysis engine** unifying Python's ast module and **tree-sitter** behind one generic analyzer interface, supporting 10 languages with deterministic **cross-file call-graph resolution**.
- Architected an optional AI chat feature fully isolated from the deterministic analysis pipeline — grounded only in already-computed call-graph results, keeping the core parsing product **100% LLM-free** with zero hallucination risk — backed by a **commit-hash-keyed PostgreSQL cache** and least-privilege, **zero-scope GitHub OAuth**.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript (ES6+), TypeScript, C, C++, SQL, HTML5, CSS3/SCSS
**Frontend:** React.js, Next.js, shadCN, Tailwind CSS, EJS
**Backend & Frameworks:** Django, Django REST Framework, Node.js, Express.js, Temporal, JWT, OAuth 2.0
**Cloud & Infrastructure:** AWS (EC2, S3, Lambda, RDS, DynamoDB, IAM, CloudWatch), Kubernetes, Vercel
**Databases:** PostgreSQL, MongoDB
**Testing & Quality Assurance:** Playwright (E2E Testing), Sentry, Visual Regression Testing, Test Automation Architecture
**DevOps & Tooling:** Docker, Docker Compose, Terraform, Git, GitHub, npm Workspaces, Postman, Linux/Bash, CI/CD

## EXTRACURRICULAR

**Technical Sciences and Movies Photography Club (TMP)**, Member (2023–2025) — organized tech events, coding contests, workshops; mentored peers.
**Omega LEO Club**, Member (2024–2025) — led community service initiatives.